Cloud Application Development Environments

AWS · Azure · Google Cloud



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Introduction to Cloud Computing

On-demand IT resources delivered over the internet, paid for as you use them

Service models

SaaS — Ready-to-use software over the web, such as webmail

PaaS — A platform to build and deploy apps, such as Elastic Beanstalk

IaaS — Virtual machines, storage and networks, such as EC2

Deployment models: Public · Private · Hybrid · Community

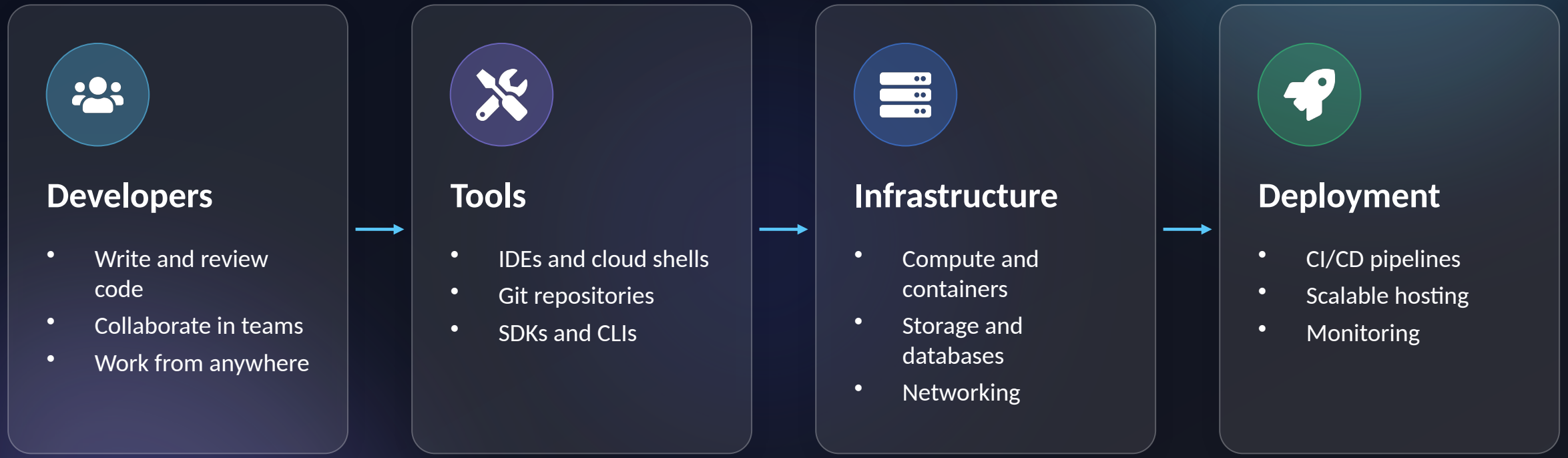
Five essential characteristics

As defined by NIST SP 800-145

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Cloud Development Environments

From a developer's keyboard to a running, scalable application



Result: faster releases, consistent environments and easy collaboration

Benefits of Cloud Development

Six reasons teams build on the cloud



Scalability

Resources grow or shrink with demand



Flexibility

Pick the languages, frameworks and services you need



Collaboration

Teams share code, environments and pipelines



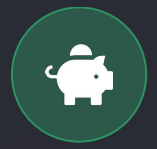
Automation

Infrastructure and releases run as code



Availability

Multiple regions and zones keep apps online



Cost efficiency

Pay only for what you use

Amazon Web Services (AWS)

The broadest portfolio of cloud services, from servers to CI/CD

Amazon EC2
Resizable virtual servers

AWS Lambda
Run code without managing servers

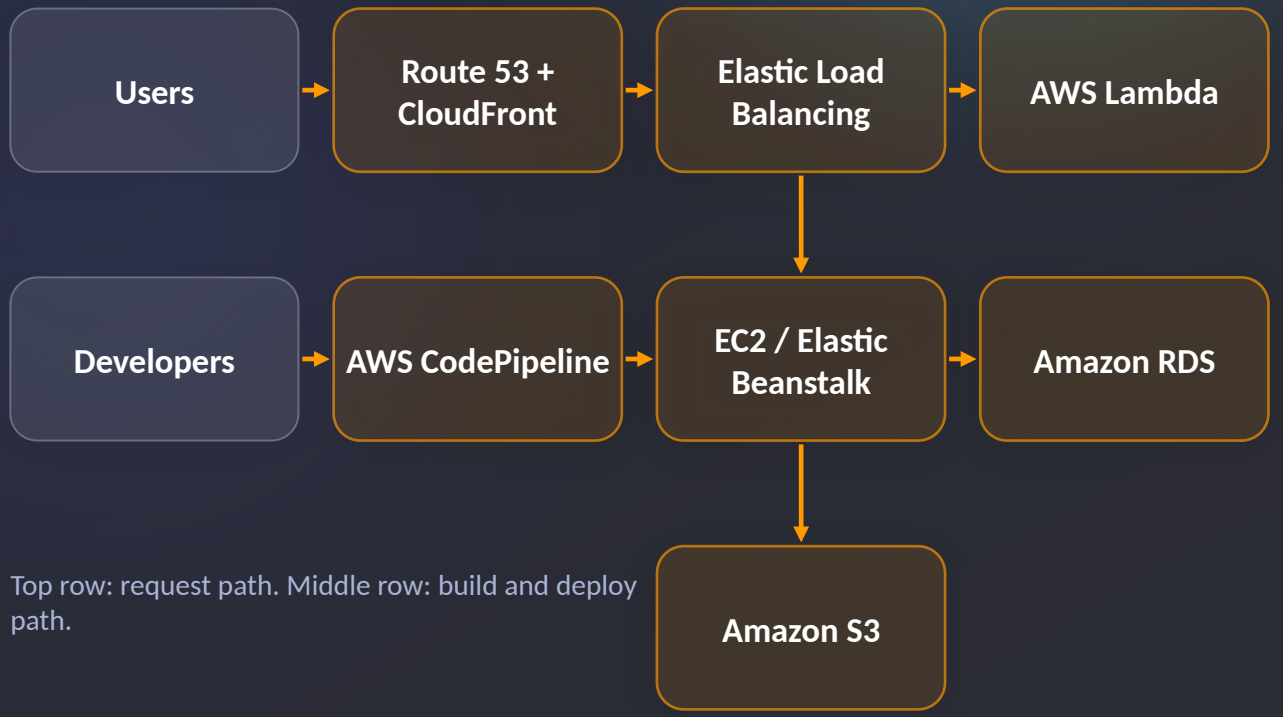
AWS Elastic Beanstalk
Deploy web apps; AWS handles the setup

Amazon S3
Scalable object storage

Amazon RDS
Managed relational databases

AWS CodePipeline
Automated CI/CD release pipelines

AWS application architecture



Top row: request path. Middle row: build and deploy path.

Microsoft Azure

Enterprise-ready cloud with deep Microsoft and hybrid integration

Virtual Machines
On-demand Windows and Linux servers

Azure Functions
Event-driven serverless code

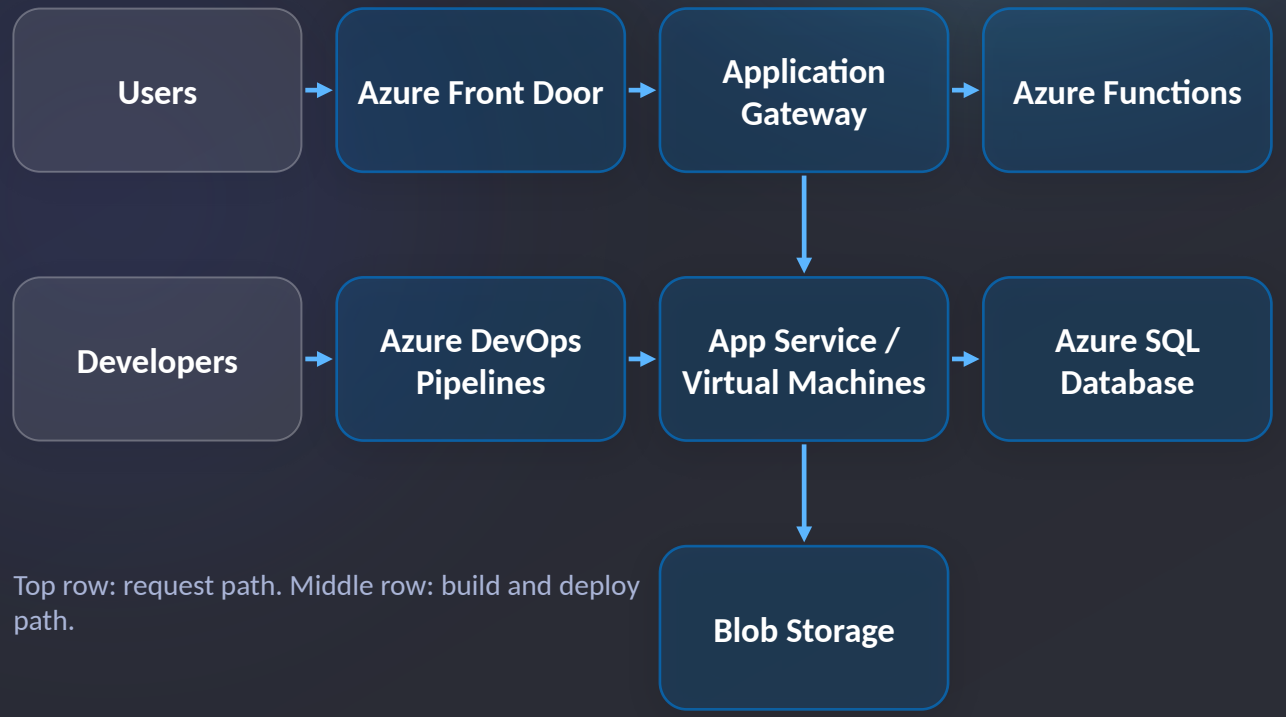
Azure App Service
Managed hosting for web apps and APIs

Blob Storage
Massively scalable object storage

Azure SQL Database
Managed relational database

Azure DevOps
Repos, pipelines and boards in one suite

Azure application architecture



Top row: request path. Middle row: build and deploy path.

Google Cloud

Strong in data, AI and container-native development

Compute Engine
Virtual machines on Google infrastructure

App Engine
Fully managed platform for web apps

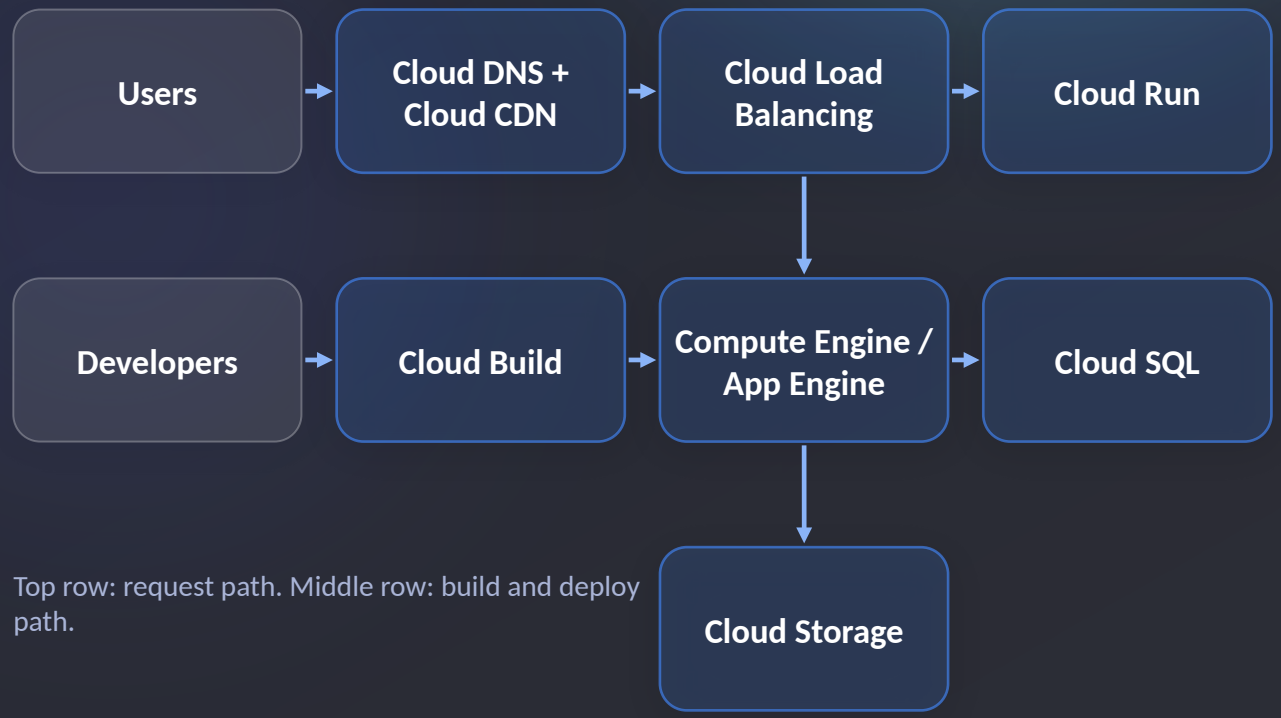
Cloud Run
Run containers without managing servers

Cloud Storage
Unified object storage

Cloud SQL
Managed MySQL, PostgreSQL, SQL Server

Cloud Build
Serverless CI/CD that builds and tests code

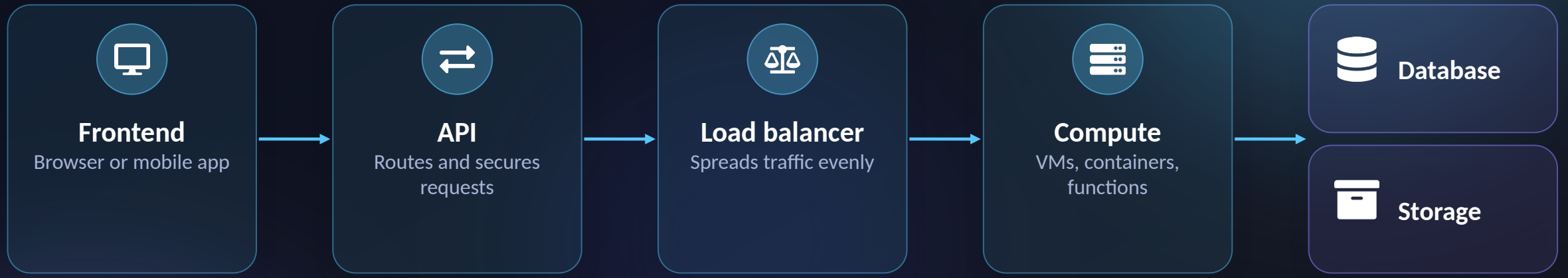
Google Cloud application architecture



Top row: request path. Middle row: build and deploy path.

Cloud Application Architecture

A cloud-native app is layered, scalable and observable

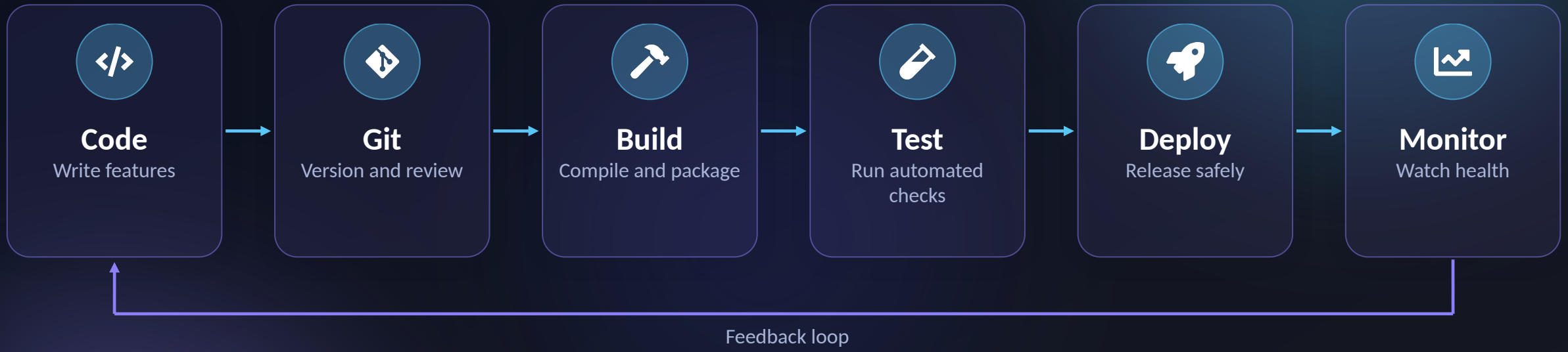


Security Identity, encryption, firewalls and network isolation protect every layer

Monitoring Metrics, logs and alerts show health and feed improvements back to the team

Development and CI/CD Workflow

Every commit flows through an automated path to production



AWS
CodePipeline · CodeBuild · CodeDeploy

Azure
Azure Repos · Azure Pipelines · Azure Monitor

Google Cloud
Cloud Build · Artifact Registry · Cloud Deploy

Compute and Hosting Comparison

Equivalent services across the three platforms

Category	AWS	Microsoft Azure	Google Cloud
Virtual machines	Amazon EC2	Virtual Machines	Compute Engine
Managed app hosting	Elastic Beanstalk	App Service	App Engine
Serverless functions	AWS Lambda	Azure Functions	Cloud Run functions
Serverless containers	AWS Fargate	Azure Container Apps	Cloud Run
Managed Kubernetes	Amazon EKS	Azure Kubernetes Service (AKS)	Google Kubernetes Engine (GKE)

Storage, Databases and Networking

Representative services for the data and network layers

Category	AWS	Microsoft Azure	Google Cloud
Object storage	Amazon S3	Blob Storage	Cloud Storage
Relational database	Amazon RDS	Azure SQL Database	Cloud SQL
NoSQL database	Amazon DynamoDB	Azure Cosmos DB	Firestore
Virtual network	Amazon VPC	Virtual Network (VNet)	VPC network
Load balancing	Elastic Load Balancing	Load Balancer / Application Gateway	Cloud Load Balancing
DNS and CDN	Route 53 + CloudFront	Azure DNS + Front Door	Cloud DNS + Cloud CDN

Security, Monitoring and DevOps

Five layers protect and operate a cloud application

	AWS	Microsoft Azure	Google Cloud
Identity and access	AWS IAM	Microsoft Entra ID	Cloud IAM
Network security	VPC · Security Groups	VNet · Network Security Groups	VPC · Firewall rules
Data encryption	AWS KMS	Azure Key Vault	Cloud KMS
Logging and monitoring	CloudWatch · CloudTrail	Azure Monitor	Cloud Logging · Monitoring
Deployment automation	AWS CodePipeline	Azure Pipelines	Cloud Build · Cloud Deploy

Advantages, Limitations and Use Cases

A balanced view: the best choice depends on the workload

AWS

ADVANTAGES

- Widest range of services
- Mature, large partner ecosystem

LIMITATIONS

- Many options can feel complex
- Costs need careful tracking

BEST FOR

- Varied, high-scale workloads

Microsoft Azure

ADVANTAGES

- Works well with Microsoft tools
- Strong hybrid cloud support

LIMITATIONS

- Portal and naming can confuse
- Some services overlap

BEST FOR

- Enterprises on .NET and Microsoft 365

Google Cloud

ADVANTAGES

- Strong data, analytics and AI
- Strong Kubernetes and containers

LIMITATIONS

- Smaller market share than rivals
- Smaller partner ecosystem

BEST FOR

- Data, AI and container-native apps

Practical Example: Student Portal

The same web app, developed, deployed, secured and monitored on each platform

	Develop	Deploy	Secure	Monitor
AWS	Git + CodePipeline CodeBuild	Elastic Beanstalk RDS + S3	IAM roles · KMS Security Groups	CloudWatch dashboards and alarms
Azure	Azure Repos Azure Pipelines	App Service Azure SQL + Blob	Microsoft Entra ID Key Vault · NSG	Azure Monitor Application Insights
Google Cloud	Git + Cloud Build Artifact Registry	Cloud Run Cloud SQL + Storage	Cloud IAM · KMS VPC firewall rules	Cloud Logging Cloud Monitoring

Conclusion, References and Thank You

Key takeaways

- Cloud environments speed up building, testing and deploying apps
- AWS, Azure and Google Cloud offer equivalent core services with different strengths
- CI/CD, security and monitoring are built into every platform
- Choose by workload, team skills and cost

Thank you

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